Coupling Risk Perception and Epidemics: A Comprehensive Framework for Co-Evolution Models

Christof Schütte[1,2]*, Thomas Gaskin[3,4], Sebastian Zimper[1], Natasa Conrad[1]

[1] Zuse Institute Berlin, Takustr. 7, 14195 Berlin, Germany; [2] Department of Mathematics and Computer Science, Freie Universität Berlin, Arnimallee 6, 14195 Berlin, Germany; [3] Department of Methodology, London School of Economics and Political Science, London WC2A 2AE, United Kingdom; [4] Data Science Institute, London School of Economics and Political Science, London WC2A 2AE

Correspondence: *schuette@zib.de

Abstract We propose a comprehensive approach to the coevolution of infection dynamics and opinion spreading, focusing on risk perception as a driver of behavioral change. The approach systematically connects agent-based models with their large-population, continuum limits. Our goal is not to optimize realism in isolated subsystems but to provide a modular, extensible framework for integrating behavioral and epidemiological models while allowing scalable simulation. We address the numerical complexities inherent in the hybrid structure of the resulting models and propose efficient simulation strategies to manage them effectively. Computational experiments illustrate key patterns in the dynamics of perception–infection feedback, showing that even rather simple models reveal rich co-evolutionary effects such as delayed epidemic peaks. Our results emphasize the need for integrated modeling of behavioral and epidemiological processes to support effective public health responses.

Keywords Epidemics, Public opinion, Co-evolution, Agent-based models, Mean-field models

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1. Introduction

Understanding the mutual feedback between human behaviour and the spread of infectious diseases remains a central challenge in modern epidemiology. In particular, the interplay between *infection dynamics* and *opinion dynamics*—including risk perception, behavioural adaptation, and trust in interventions—has gained increasing relevance in recent years [1, 2]. While each of these processes has been extensively studied in isolation, there is a fundamental need for integrated *co-evolution models* that capture their bidirectional coupling. The COVID-19 pandemic has highlighted phenomena such as the rise of infodemics, the polarisation of opinions on vaccines, and the dissemination of conspiracy theories [3, 4], with similar concerns arising for other infectious diseases [5].

In this paper, we propose a comprehensive modelling framework based on agent-based models (ABMs) in both continuous and discrete opinion spaces. ABMs simulate the actions and interactions of individuals represented as agents, capturing heterogeneity and complex contact structures. They have become essential tools for studying infectious disease dynamics as well as the dissemination of opinions and beliefs. We show that the structure of ABMs allows for straightforward construction of co-evolution models by coupling existing models at their respective levels. Furthermore, we demonstrate how to transition from ABMs to macro-models by deriving the corresponding large-population (mean-field) limits and then constructing fully coupled co-evolution models linking opinion states to infection risks and vice versa.

Our approach is *not* intended to incorporate the most detailed or advanced models of infection, opinion, or social behaviour. Rather, it provides a clear and flexible pathway for systematically coupling such models, enabling researchers to extend existing infection or opinion dynamics models into coherent co-evolution frameworks at micro- (ABMs), meso-, and macro-levels (mean-field). While several studies have addressed transferring coupled models from micro- to macro-scales [6, 7], these are often restricted to specific examples. In contrast, our framework is general, offering a systematic approach for co-evolution modelling rather than focusing on a particular multiscale or aggregation method.

We emphasise the importance of incorporating empirical approaches from the social and public health sciences to enhance applicability to real-world scenarios [3]. Throughout the paper, we provide remarks indicating potential extensions, for instance regarding vaccination hesitancy and disease spread [4, 8, 9].

To efficiently simulate these complex dynamics, we employ numerical methods including operator splitting schemes and hybrid particle–PDE algorithms, which manage the high-dimensional and nonlinear structure of the coupled systems. We present a series of numerical experiments illustrating typical scenarios of opinion–infection co-evolution. These include settings with time-varying risk perception, bounded confidence in social interactions, and asymmetric switching behaviour. The results demonstrate how feedback between concern levels and infection prevalence can drive nonlinear transitions, including multiple infection waves, even when based on a simple SIR model [10]. This is significant, as standard SIR-type models are known to be incapable of capturing phenomena such as multiple waves or infection peaks [1, 2, 11]. Finally, we calibrate our model to real data of infection and social opinion from Japan during the early phases of the COVID pandemic, demonstrating the applicability of this framework to observable phenomena.

Our work contributes to the growing literature on coupled behavioural–epidemiological models [3, 6, 8, 11–13], offering a mathematically rigorous and computationally tractable approach. By starting from ABMs, which have previously integrated infection dynamics with aspects such as activity patterns in real-world settings [14], we show how to extend these models to macro-levels suitable for efficient simulation, parameter estimation, and optimal control strategies [15]. This lays the groundwork for future extensions incorporating realistic interaction networks, data-informed parameterisation, and learning-based calibration.

The remainder of the paper is structured as follows.

Section 2 introduces agent-based models for infection dynamics and their large-population limits. Section 3 presents ABMs for discrete and continuous opinion dynamics alongside their corresponding continuum descriptions. Section 4 develops a systematic framework for coupling the two systems into coherent co-evolution models. Section 5 addresses numerical challenges and introduces efficient simulation strategies. Section 6 illustrates the dynamical behaviour of the coupled models through representative numerical experiments for discrete and continuous opinion spaces, respectively, as well calibrating the continuum co-evolution model to real data. Section 8 concludes with a summary and discussion of future directions.

2. Infection dynamics

Models of infectious disease spread can be broadly classified into *agent-based models* (ABMs), *compartmental models*, and *PDE-type models*. Agent-based models simulate the actions and interactions of individual agents, thereby capturing heterogeneity and complex contact structures. Compartmental models aggregate individuals into population classes (e.g., SIR) and describe transitions between them through systems of differential equations. PDE-type models extend compartmental approaches by incorporating spatial structure and transport, enabling the study of travelling waves and spatial heterogeneity. In the following, we outline how these model types are interrelated.

2.1. ABMs for infection spreading

ABMs have become indispensable tools in infectious disease research, particularly when heterogeneity in individual behaviour, contact networks, or mobility patterns plays a decisive role in shaping epidemic outcomes [15, 16]. In contrast to compartmental models (e.g. SIR [10]), which in their classical ODE formulation describe population-level averages [17], ABMs explicitly simulate discrete individuals (*agents*) embedded in a social or spatial environment. Each agent follows behavioural rules that may depend on demographic attributes, mobility, health status, and responses to public health interventions.

This bottom-up perspective allows the inclusion of features—such as heterogeneous mixing, adaptive behaviour, network effects, and realistic geography—that are difficult or impossible to represent in aggregated models. In epidemiology, ABMs have been used to evaluate intervention strategies, assess vaccination policies, and study the co-evolution of pathogen transmission and behavioural adaptation [14, 18–22]. By explicitly representing individuals and their interactions, ABMs provide a flexible and interpretable framework that bridges epidemiology, social science, and computational modelling, making them particularly suitable for data-driven and policy-relevant analyses.

Let us first focus only on an ABM representing infection dynamics, abstracting from additional components such as spatial structure or activity patterns, which can readily be incorporated later. Such ABMs for infection dynamics may operate in either discrete or continuous time. In discrete-time models, all agents update synchronously at fixed time steps, whereas in continuous time, events such as infection or recovery occur asynchronously according to stochastic rates. Discrete-time schemes are often simpler to implement and align naturally with data reported in regular intervals, while continuous-time formulations capture event timing more precisely and avoid artefacts introduced by fixed-step updates. The choice depends on the intended balance between realism, computational tractability, and data compatibility. Here, we focus on continuous-time ABMs.

Prototypical infection ABM In the most basic formulation, each agent occupies one of M infection states, denoted by $Y_k(t) \in \{1, \dots, M\}$. Agents in infection state Y_i are referred to as being of type $\mathscr{S}_i$. For example, in an ABM with the classical SIR structure, $M = 3$, corresponding to $\mathscr{S}_1 = S$ ("susceptible"), $\mathscr{S}_2 = I$ ("infected"), and $\mathscr{S}_3 = R$ ("recovered"). In an SEIR model, an additional E ("exposed") state is included, while an SEIRD model further distinguishes a "deceased" state. If multiple pathogen variants are present, the E and I types can be duplicated accordingly.

Agents change their infection states through *infection events*. Each event represents a *transition* of a specific form, and the model comprises K distinct transition types. Every transition type $\mathscr{R}_k$, $k = 1, \dots, K$, is

represented by a rule of the form

$$\mathscr{R}_k: \quad a_{1k}\mathscr{S}_1 + \cdots + a_{Mk}\mathscr{S}_M \to b_{1k}\mathscr{S}_1 + \cdots + b_{Mk}\mathscr{S}_M, \quad (1)$$

where $\mathscr{S}_i$ denotes agents of infection state i, and the coefficients $a_{ik}, b_{ik} \in \mathbb{N}_0$ specify the numbers of agents of each type participating in the transition as input and output, respectively. To ensure conservation of the total number of agents, we assume

$$\sum_{i=1}^{M} a_{ik} = \sum_{i=1}^{M} b_{ik} \; \forall k.$$

The *associated vector* $\boldsymbol{\nu}_k = (\nu_{1k}, \ldots, \nu_{Mk}) \in \mathbb{Z}^M$ is defined by

$$\nu_{ik} := b_{ik} - a_{ik},$$

and describes the *net change* in the number of agents of each type $\mathscr{S}_i$ resulting from transition $\mathscr{R}_k$. Different transitions may produce identical net changes, corresponding, for instance, to distinct social mechanisms inducing the same overall effect.

The *population state* (or *system state*) of the ABM is represented by the vector

$$\tilde{y} = (\tilde{y}_1, \ldots, \tilde{y}_M) \in \mathbb{N}_0^M,$$

where $\tilde{y}_i$ denotes the number of agents of type $\mathscr{S}_i$, i.e. those currently in infection state i.

The stochastic process $Y(t)$ for $t \geq 0$ is a continuous-time Markov jump process: it remains piecewise constant, with random jumps of the form

$$\tilde{y}(t) \to \tilde{y}(t) + \boldsymbol{\nu}_k,$$

each corresponding to an active transition $\mathscr{R}_k$. The waiting times between jumps are exponentially distributed, with rates determined by the *propensity functions* α_k, which quantify the probability per unit time that transition $\mathscr{R}_k$ occurs given the current state of the system [23]. The system dynamics are thus given by

$$\tilde{y}(t) = \tilde{y}(0) + \sum_{k=1}^{K} U_k\left(\int_0^t \alpha_k(s)\,\mathrm{d}s\right)\boldsymbol{\nu}_k, \quad (2)$$

where U_k, $k = 1, \ldots, K$, are independent, unit-rate Poisson processes.

Let

$$P(\tilde{y}, t) := \mathbb{P}(\tilde{y}(t) = \tilde{y} \mid \tilde{y}(0) = \tilde{y}_0)$$

denote the probability of finding the system in state $\tilde{y}$ at time t, given the initial condition $\tilde{y}_0$. The time evolution of this probability is governed by the *chemical master equation* [24]:

$$\frac{\mathrm{d}P(\tilde{y}, t)}{\mathrm{d}t} = \sum_{k=1}^{K}\Big[\alpha_k(\tilde{y} - \boldsymbol{\nu}_k)P(\tilde{y} - \boldsymbol{\nu}_k, t) - \alpha_k(\tilde{y})P(\tilde{y}, t)\Big], \quad (3)$$

where $\alpha_k(\tilde{y}) := 0$ and $P(\tilde{y}, t) := 0$ for $\tilde{y} \notin \mathbb{N}_0^M$, ensuring that the right-hand side of eq. (3) excludes terms with negative entries in $\tilde{y} - \boldsymbol{\nu}_k$.

Remark 2.1. Many ABMs are defined by a set of interaction rules, with each agent iteratively or randomly subjected to one of the applicable rules. This is a fully agent-oriented formulation that however leads to an exponentially large state space $\{1, \ldots, M\}^N$ for large populations. As an alternative, we describe the system by its *population state* $\tilde{y}$, which records the number of agents of each type. For homogeneous agents and a complete interaction network—or equivalently, for randomly interacting agents—this population-level description is exact; otherwise, it introduces an approximation through aggregation. In what follows, we assume complete mixing for simplicity. Under this assumption, the transition rules between agent types translate directly into transition rules between population states as outlined above. In Sec. 4.3, however, we also consider a fully agent-oriented formulation for opinion dynamics.

Example 2.2 (SIR example)**.** Throughout this paper, we use the *SIR model* as a guiding example. In this setting, agents can occupy three infection states ($M = 3$): $\mathscr{S}_1 = S$, $\mathscr{S}_2 = I$, and $\mathscr{S}_3 = R$. Two types of transitions are considered: first, *infection transitions* are of the form

$$\mathscr{R}_1: \quad \mathscr{S}_1 + \mathscr{S}_2 \to 2\mathscr{S}_2,$$

with $\boldsymbol{\nu}_1 = (-1, 1, 0)^\top$, or equivalently, in standard SIR notation,

$$\mathscr{R}_1: \quad S + I \to 2I.$$

This describes the infection of a susceptible agent upon contact with an infected one.

Recovery transitions are given by

$$\mathscr{R}_2: \quad \mathscr{S}_2 \to \mathscr{S}_3,$$

with $\boldsymbol{\nu}_2 = (0, -1, 1)^\top$, or in SIR notation,

$$\mathscr{R}_2: \quad I \to R,$$

representing the recovery of an infected agent independently of others.

The corresponding propensity functions are

$$\alpha_1(\tilde{y}) = \frac{k_I}{N}\,\tilde{y}_1\tilde{y}_2, \qquad \alpha_2(\tilde{y}) = k_R\tilde{y}_2,$$

where $k_I \geq 0$ and $k_R \geq 0$ denote the infection and recovery rate constants, respectively. The $1/N$ scaling in α_1 accounts for the probability that two agents meet in a well-mixed population of size N.

Most existing ABMs for infection spreading already allow for the integration of non-pharmaceutical interventions, such as social distancing and mask mandates, while also including pharmaceutical measures like vaccination campaigns to assess their impact on disease spread and public health outcomes, as well as spatial spreading [14].

2.2. Population limit

For large N, an appropriate rescaling of the dynamics leads to effective approximate dynamics given by Fokker-Planck, stochastic, or ordinary differential equations. In what follows, we summarize the corresponding results [23, 25] which are mainly based on the law of large numbers and the central limit theorem. The system state for a system of total population N is now denoted by $\tilde{y}^N$; for the propensity functions we now also use the notation α_k^N in order to indicate their dependence on N. We introduce the relative frequencies $y = \tilde{y}/N$ together with the *scaled propensities*

$$\tilde{\alpha}_k(y) := N^{-1}\alpha_k^N(yN),$$

where the scaling is correct for up to second order transitions and reflects that, with increasing N and without taking into account spatial distributions, the propensities have to be scaled in order to reflect the probabilities for the underlying transition in a well-mixed ensemble, compare [24].

It is well-known that in the limit $N \to \infty$ and assuming convergence of the scaled propensity functions, the rescaled jump process $\tilde{y}^N(t)/N|_{t\geq 0}$ converges to the frequency process $y(t)|_{t\geq 0}$ given by the ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}y(t) = \sum_{k=1}^{K} \tilde{\alpha}_k(y(t)) \cdot \nu_k \tag{4}$$

with initial state $y(0) = \lim_{N\to\infty} \tilde{y}^N(0)/N$.

The associated Fokker–Planck or Liouville equation has the form

$$\frac{\partial}{\partial t}\rho(y,t) = -\sum_{i=1}^{m} \frac{\partial}{\partial y_i}\Big(b_i(y)\cdot\rho(y,t)\Big),$$

that is,

$$\partial_t\rho = -\mathrm{div}_y(b\cdot\rho)$$

with

$$b(y) := \sum_{k=1}^{K} \tilde{\alpha}_k(y)\cdot\nu_k,$$

where $y = (y_1, \ldots, y_m)$ and the $y_k \in [0,1]$ now denotes the fraction of population being in infection state $k \in \{1, \ldots, m\}$. That is, eq. (2.2) is a PDE for a scalar density ρ that depends on m dimensions.

Let Φ^t denote the deterministic flow given by eq. (4), that is, $\Phi^t y(0)$ is the solution of eq. (4) at time t when started in $y(0)$. Then,

$$\rho(y,t) = \rho_0\Big(\Phi^{-t}y\Big) \tag{5}$$

is the solution of (2.2) with initial condition $\rho(y, t = 0) = \rho_0(y)$.

Example 2.3 (SIR example continued)**.** For the simple SIR model, the concentrations $s = S/N = y_1/N$, $i = I/N = y_2/N$, $r = R/N = y_3/N$ satisfy a Fokker–Planck (or Liouville) equation describing transport along the trajectories of the deterministic ODE system derived from eqs. (5) and (4). With $C = (s, i, r)$ and

$$b(C) = (-k_I si,\ k_I si - k_R i,\ k_R i),$$

the corresponding system reads

$$\frac{\mathrm{d}s}{\mathrm{d}t}(t) = -k_I\, s(t)\, i(t), \tag{6}$$

$$\frac{\mathrm{d}i}{\mathrm{d}t}(t) = k_I\, s(t)\, i(t) - k_R\, i(t), \tag{7}$$

$$\frac{\mathrm{d}r}{\mathrm{d}t}(t) = k_R\, i(t), \tag{8}$$

which is the classical compartmental form of the SIR model.

In the standard interpretation,

$$\tau_R := 1/k_R$$

is the mean infectious period, i.e. the average time between infection and recovery for an individual. The ratio

$$\mathcal{R}_0 := k_I/k_R$$

is the *basic reproduction number*, representing the expected number of secondary infections caused by one infectious individual during the full infectious period τ_R in a fully susceptible population ($s \approx 1$). The infection spreads as long as the susceptible fraction s exceeds the critical value

$$s_{\text{crit}} = 1/\mathcal{R}_0.$$

Remark 2.4. The classical SIR-equations (6)–(8) correctly capture the first-order moment of the convergence in N of the distributions of the infection types resulting from the underlying ABM. In order to also approximate the second moment well, one has to consider the "chemical Langevin" form of the compartment SIR dynamics,

$$\begin{aligned}\mathrm{d}y(t) = &\sum_{k=1}^{K} \tilde{\alpha}_k(y(t)) \cdot \nu_k \mathrm{d}t \\ &+ \sum_{k=1}^{K} \frac{1}{N}\sqrt{\tilde{\alpha}_k(y(t))}\, \mathrm{d}B_k(t)\, \nu_k,\end{aligned} \tag{9}$$

with independent Brownian motions B_k: see [26] or [24], Sec. 2.1.3.

3. Opinion dynamics

Models of opinion dynamics can be formulated using either *discrete* or *continuous* opinion states. In discrete models, agents occupy one of a finite set of opinions (e.g., "concerned" or "unconcerned"), while in continuous models, opinions vary smoothly along a spectrum (e.g., from "fully unconcerned" to "extremely concerned"). Discrete formulations are typically simpler and naturally align with compartmental frameworks, whereas continuous ones provide finer resolution and capture gradual shifts and heterogeneity in attitudes. The choice between them depends on the level of behavioural detail desired and the associated computational cost.

In both discrete and continuous ABMs, social relations—such as friendships or online connections—are represented by a (possibly time-dependent) adjacency matrix

$$A(t) \in \{0,1\}^{N\times N}.$$

An entry $A_{ij}(t) = 1$ indicates that agents i and j can interact at time t, while $A_{ij}(t) = 0$ indicates the absence of a link. When the network is weighted or directed, i.e. $A(t) \in [0,1]^{N\times N}$, the matrix can also encode the strength or direction of social influence. Such matrices can be inferred, for example, from online social network data or sociological surveys [27].

For simplicity, we assume in the following that A corresponds to a complete network, with $A_{ij} = 1$ for all $i \neq j$ and $A_{ii} = 0$. Extensions to general network topologies are straightforward.

3.1. Discrete opinion dynamics

Our starting point is a continuous-time opinion dynamics ABM where agents are nodes in an interaction network, similar to the discrete-time model given in [28].

Prototypical opinion ABM Each agent can switch according to given transition rules between finitely many states (called *types*), comparable to the infection dynamics ABM considered in Sec. 2. We allow the agent to select between M' different opinions so that the type space is $\{1,\dots,M'\}$, then for our N agents the state space of the ABM the configuration space contains M'^N different configurations, i.e. its size grows combinatorially like M'^N, which poses a problem when N becomes very large. Like in the infection case, an agent k being of opinion state $X_k(t) = i$ at time t is said to be of type $\mathscr{S}_i'$, and

the transition rules $\mathscr{R}'_k$, $k = 1, \ldots, K'$, are represented by an equation of the form

$$\mathscr{R}'_k : a'_{1k}\mathscr{S}'_1 + \ldots + a'_{M'k}\mathscr{S}'_m \to b'_{1k}\mathscr{S}'_1 + \ldots + b'_{M'k}\mathscr{S}'_m,$$

with associated change vector $\nu'_{ik} := b'_{ik} - a'_{ik}$ and propensity functions $\alpha'_{ik}(t)$. The dynamics of the ABM is again given by a Markov jump process whose evolution in time is, as in (2), given by

$$X^N(t) = X^N(0) + \sum_{k=1}^{K'} U'_k\Big(\int_0^t \alpha'_k(s)\mathrm{d}s\Big)\,\nu'_k,$$

where the N-superscript again refers to the number of agents in the system.

3.2. Population limit

In complete analogy to Sec. 2, after re-scaling of the propensity functions, the rescaled jump process $X^N(t)/N|_{t\geq 0}$ converges in the limit $N \to \infty$ to the frequency process $x(t)|_{t\geq 0}$ given by the ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = \sum_{k=1}^{K'} \tilde{\alpha}'_k(x(t)) \cdot \nu'_k$$

with initial state $x(0) = \lim_{N\to\infty} X^N(0)/N$. The associated Fokker-Planck or Liouville equation takes the form

$$\partial_t \rho = -\mathrm{div}_x(b' \cdot \rho)$$

with $b'(x) := \sum_{k=1}^{K'} \tilde{\alpha}'_k(x) \cdot \nu'_k$, where $x = (x_1, \ldots, x_n)$ now denotes the fraction of population being in opinion states $k \in \{1, \ldots, n\}$.

Example 3.1 (Binary opinion space)**.** Consider the simplest possible example of a binary opinion space with just two states ($M' = 2$), unconcerned (0) and concerned (+1) and two reactions, so that $K' = 2$; the opinion dynamics are then described by just two transition rules,

$$\mathscr{R}'_1 : \quad \mathscr{S}'_0 \to \mathscr{S}'_1,$$

with $\nu'_0 = (-1, 1)^\top$, $\alpha'_0(y) = a_0 x_0$ and

$$\mathscr{R}'_1 : \quad \mathscr{S}'_1 \to \mathscr{S}'_0,$$

with $\nu'_1 = (1, -1)^\top$, $\alpha'_1(y) = a_1 x_1$. The associated "mean-field" Liouville equation for this opinion ABM in the population limit is given by

$$\partial_t \rho = -\mathrm{div}_x(b' \cdot \rho)$$

with

$$b'(x) = (-a_0 x_0 + a_1 x_1, a_0 x_0 - a_1 x_1)^\top$$

and associated linear ODE system

$$\frac{\mathrm{d}}{\mathrm{d}t}x_0 = -a_0 x_0 + a_1 x_1$$
$$\frac{\mathrm{d}}{\mathrm{d}t}x_1 = a_0 x_0 - a_1 x_1.$$

3.3. Continuous opinion dynamics

We consider N agents with opinions

$$\boldsymbol{X}(t) = (X_i(t))_{i=1,\ldots,N} \in \mathbb{O}^N,$$

where each $X_i(t)$ lies in a *continuous* d-dimensional opinion space $\mathbb{O} \subset \mathbb{R}^d$. For example, when modelling opinions related to epidemic risk perception, a one-dimensional space ($d = 1$) may represent the level of concern, ranging continuously from "fully unconcerned" to "extremely concerned" or even "panicked". If an additional attitude—such as vaccination hesitancy—is included, the opinion space becomes two-dimensional ($d = 2$), with one axis describing perceived risk and the other spanning from "strongly pro-vaccination" to "strongly anti-vaccination". Throughout this section, we consider a simple continuous and bounded opinion space, such as $\mathbb{O} = [-1, 1]$, while noting that higher-dimensional extensions follow naturally.

Prototypical ABM As our basic ABM for opinion dynamics we consider the well-known Hegselmann-Krause model [29] with generalizations like in [30, 31].

We again consider N agents moving in $\mathbb{O}$ with no-flux or periodic boundary conditions. Let $X(t) = (X_1(t), \ldots, X_N(t))$ denote the (random) system state in $\mathbb{O}$ at time $t \in [0, \infty)$, where $X_k(t) \in \mathbb{O}$ is the opinion of agent $k \in \{1, \ldots, N\}$.

First, we consider an opinion dynamics ABM without coupling to the infection dynamics. The opinions evolve according to the set of coupled SDEs

$$\begin{aligned} dX_k(t) = &- \nabla V(X_k(t),t)dt \\ &- \frac{1}{N}\sum_{j=1,j\neq i}^{N} \nabla W(X_k(t)-X_j(t))\,dt \\ &+ \sigma dB_k(t), \end{aligned} \tag{10}$$

where V and W denote the external and the interaction potentials, respectively, and the B_k independent Brownian motions, respectively. The external potential V is a smooth function on $\mathbb{O}$; for simplicity, we assume that it grows so strongly towards the boundary of $\mathbb{O}$ that touching the boundary is extremely unlikely and can be ignored. For the interaction potential W, we here assume that it is one of the potentials used in the context of bounded confidence modeling [29, 32] that only allows interactions with other individuals that are within a radius d.

Remark 3.2. In some models, the interaction term $\nabla W(X_k(t) - X_j(t))$ is replaced with a more general kernel interaction of the form $\mathscr{K}(X_k(t), X_j(t))$ with a smooth interaction kernel $\mathscr{K}$. Moreover, when the adjacency matrix $A(t)$ of the interaction network between the agents is not full, then the sum in (10) is replaced by a sum over all agents $j \neq i$ with $A_{ij}(t) > 0$.

3.3.1 Population limit / mean-field limit

For large population sizes N, the particle-based dynamics may be approximated by a mean-field PDE, compare [33]. This equation describes the spatiotemporal evolution of the density $\rho(x,t)$ of particles, depending on opinion position $x \in \mathbb{O}$ and time $t \geq 0$, and approximates the empirical measure of the particle-based dynamics in the case of large N. The associated *mean-field PDE* is given by:

$$\begin{aligned} \partial_t \rho &= \nabla_x \cdot \Big(\rho \cdot (\nabla_x V + \nabla_x W * \rho)\Big) + \frac{\sigma^2}{2}\Delta_x \rho \\ &= \mathrm{div}_x\Big(\rho \cdot (\nabla_x V + \nabla_x W * \rho)\Big) + \frac{\sigma^2}{2}\Delta_x \rho, \end{aligned} \tag{11}$$

where $*$ denotes convolution between the interaction force $\nabla_x W$ and the density ρ. This partial integro-differential equation is a nonlinear, non-local Fokker-Planck equation that is also called *McKean-Vlasov PDE* or *aggregation-diffusion equation*.

Furthermore, the associated mean-field SDE describes the evolution of a *single* typical agent $X = X(t)$ in the mean-field density field $\rho = \rho(t,x)$:

$$\begin{aligned} dX(t) = &- \nabla V(X(t),t)dt \\ &- (\nabla W * \rho)(X(t),t)\,dt + \sigma dB(t). \end{aligned} \tag{12}$$

Remark 3.3. The mean-field PDE (11)) approximates the particle-based dynamics for $N \to \infty$, but it lacks stochastic effects crucial for systems of finite size N, especially in the case of clustering behavior. This randomness is maintained by the corresponding *stochastic partial differential equation (SPDE)*

$$\begin{aligned} \partial_t \rho = &\ \mathrm{div}_x\Big(\rho \cdot (\nabla_x V + \nabla_x W * \rho)\Big) \\ &+ \frac{\sigma^2}{2}\Delta_x \rho + \frac{\sigma}{\sqrt{N}}\mathrm{div}_x(\sqrt{\rho}\xi), \end{aligned} \tag{13}$$

also called *Dean–Kawasaki equation* [34–36], where $\xi = \xi(x,t)$ denotes space-time white noise.

For the mean-field PDE (11, the period of cluster formation has been studied analytically based on linear stability analysis [37], providing estimates for the number of clusters and the time to cluster formation, showing phase transitions in dependence on the strength of the interaction potential W. Once clusters have formed, their total number and individual positions are an invariant of the deterministic system of motion. To continue with *clustering dynamics* including growth, birth and death of clusters, further stochastic effects are required. In [38], it is demonstrated that this long-term stochastic clustering dynamics is described by the SPDE (13) is in very close alignment with that of the particle-based model.

4. Co-evolution dynamics

4.1. Discrete opinion space

If the opinion space is discrete, the opinion ABM and the infection ABM, both, take the form of a set of transition

rules for the switching between (discrete) states that can directly be compiled into a single set-up with $K' + K$ transitions. The complete coupled system can then again be written as a reaction system in $M \times M'$ states (see Example 4.1). For N agents, the configuration space is of size $(M \times M')^N$ and again grows combinatorially for large populations, an effect that makes mean-field models very attractive for concrete simulations. The associated mean-field PDE has the general form

$$\begin{aligned}\partial_t \rho(x,y,t) = &- \operatorname{div}_x(b'(x,y)\cdot\rho(x,y,t)) \\ &- \operatorname{div}_y(b(x,y)\cdot\rho(x,y,t))\ , \end{aligned} \tag{14}$$

with b and b' as introduced above.

However, for the co-evolution ABM as well as the associated means-field PDE, we are now free to allow for an insertion of additional transition rules, or for a dependence of the kinetic rates in the respective propensity functions on the respective "opposite" states. That is, the rates k_i in the propensities composing b of the infection ABM now are allowed to depend on x, while the rate in the propensities of the opinion ABM that compose b' may depend on y.

Example 4.1 (Binary opinion space and SIR infection dynamics)**.** Consider the simplest possible example of a binary opinion space with just two states, unconcerned (0) and concerned (+1), Example 3.1, in combination with the SIR-like infection dynamics as described in Examples 2.2 and 2.3. For the associated co-evolution ABM the agents thus can be of five types: S_0: susceptible and unconcerned, I_0: infected and unconcerned, S_1: susceptible and concerned, I_1: infected and concerned, and R: recovered, while $\tilde{y}_1, \ldots, \tilde{y}_5$ denote the numbers of agents of types S_0, I_0, S_1, I_1, R. We have 8 transitions and associated propensities:

$$\begin{array}{lll}
\mathscr{R}_1: & S_0 + I_0 \to 2I_2 & \alpha_1 = k_{I0}\,\tilde{y}_1\tilde{y}_2/N \qquad \nu_1 = (-1,1,0,0,0) \\
\mathscr{R}_2: & S_1 + I_1 \to 2I_1 & \alpha_2 = k_{I1}\,\tilde{y}_2\tilde{y}_4/N \qquad \nu_2 = (0,0,-1,1,0) \\
\mathscr{R}_3: & I_0 \to R & \alpha_3 = k_R\tilde{y}_2 \qquad \nu_3 = (0,-1,0,0,1) \\
\mathscr{R}_4: & I_1 \to R & \alpha_4 = k_R\tilde{y}_4 \qquad \nu_4 = (0,0,0,-1,1) \\
\mathscr{R}_5: & S_0 \to S_1 & \alpha_5 = a_0\tilde{y}_1 \qquad \nu_5 = (-1,0,1,0,0) \\
\mathscr{R}_6: & S_1 \to S_0 & \alpha_6 = a_1\tilde{y}_3 \qquad \nu_1 = (1,0,-1,0,0) \\
\mathscr{R}_7: & I_0 \to I_1 & \alpha_7 = a_0\tilde{y}_2 \qquad \nu_1 = (0,-1,0,1,0) \\
\mathscr{R}_8: & I_1 \to I_0 & \alpha_8 = a_1\tilde{y}_4 \qquad \nu_1 = (0,1,0,-1,0),
\end{array}$$

where the kinetic rates $k_{I0}, k_{I1}, k_R, a_0, a_1$ may depend on the system's state: k_{I0}, k_{I1}, k_R on opinion states and a_0, a_1 on infection states, see below. Transitions that simultaneously cross infection states and opinion states, such as $(S,0) \to (I,1)$, do not appear since they are second order.

For the associated mean-field co-evolution model the relevant states are $x = (x_0, x_1)$ and $y = (y_1, y_2, y_3) = (s, i, r)$, and the evolution of the mean-field PDE (14) occurs by transport along the trajectories of a 6-dimensional ODE system, with states $(x,y) = (x_0,s),(x_1,s),(x_0,i),(x_1,i),(x_0,r),(x_1,r)$ that we rename $s_0, s_1, i_0, i_1, r_0, r_1$. Then, the ODE reads

$$\begin{aligned}
\frac{ds_0}{dt}(t) &= -k_{I0}(x)s_0(t)i_0(t) - a_0(y)s_0 + a_1(y)s_1 \\
\frac{di_0}{dt}(t) &= k_{I0}(x)s_0(t)i_0(t) - k_{R0}i_0(t) - a_0(y)i_0 + a_1(y)i_1 \\
\frac{dr_0}{dt}(t) &= k_{R_0}i_0(t) - a_0(y)r_0 + a_1(y)r_1 \\
\frac{ds_1}{dt}(t) &= -k_{I1}(x)s_1(t)i_1(t) + a_0(y)s_0 - a_1(y)s_1 \\
\frac{di_1}{dt}(t) &= k_{I1}(x)s_1(t)i_1(t) - k_{R1}i_1(t) + a_0(y)i_0 - a_1(y)i_1 \\
\frac{dr_1}{dt}(t) &= k_{R_1}i_1(t) + a_0(y)r_0 - a_1(y)r_1,
\end{aligned} \tag{15}$$

where the infection rates $(k_{I0}, k_{I_1}, k_{R_0}, k_{R_1})$ may depend on opinion states, e.g., on the present fraction of concerned, $i_1(t) + s_1(t)$, and the opinion transition rates a_0 and a_1 on y, e.g., on the present total fraction of infections, $i_0(t) + i_1(t)$. Moreover, we can combine

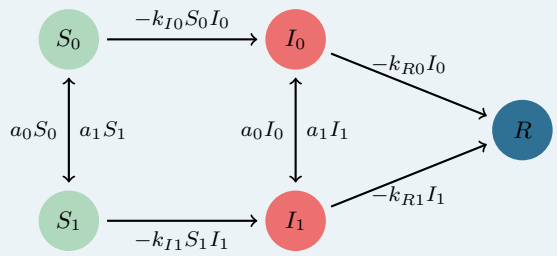

Figure 1: 5-dimensional coevolution model with discrete opinion space $\{0,1\}$ and SIR like infection dynamics. Arrows indicate the transitions of the model with formulae giving the respective propensity functions.

the two states r_0 and r_1 into one, $r = r_0 + r_1$, with ODE

$$\frac{dr}{dt}(t) = k_{R_0} i_0(t) + k_{R_1} i_1(t),$$

that is, we get a final 5-dim ODE system for the large population limit, as illustrated in Fig. 1.

4.2. Continuous opinion space

In analogy to the discrete case, the co-evolution model with continuous opinion space simply result from combining the respective ABMs or mean-field models.

4.3. Co-evolution ABM

The co-evolution ABM results from a combination of the Markov jump process ABM eq. (2) for infections and the SDE-based ABM for the opinion states eq. (10),

$$\begin{aligned}
\tilde{y}(t) &= \tilde{y}(0) + \sum_{k=1}^{K} U_k\Big(\int_0^t \alpha_k(x(s),s)ds\Big)\,\nu_k,\\
dx_k(t) &= -\nabla V(x_k(t),\tilde{y}(t))dt\\
&- \frac{1}{N}\sum_{j=1}^{N} \nabla W\left(x_k(t) - x_j(t), \tilde{y}(t)\right) dt + \sigma dB_k(t),
\end{aligned}$$

where we again assume that the kinetic rates, or, and the external and interaction potentials of the two subsystems may additionally depend on the state of the other component. Remarks on numerical simulation of such coupled processes will be given in Sec. 5. The resulting ABM lives in an $(Nd+M)$-dimensional configuration space.

4.4. Population limit / mean field PDE

In the mean-field case $N \to \infty$, we now get a population density $\rho = (x,y,t)$ that depends on the opinion and infection states simultaneously, and obeys the mean-field PDE:

$$\begin{aligned}
&\partial_t \rho(x,y,t) =\\
&\mathrm{div}_x\Big(\rho(x,y,t)\cdot(\nabla_x V(x,y) + \nabla_x W(x,y) * \rho(x,y,t))\Big)\\
&- \mathrm{div}_y(b(x,y)\cdot\rho(x,y,t)) \;+\; \frac{\sigma^2}{2}\Delta_x \rho(x,y,t),
\end{aligned} \tag{16}$$

with derivation in analogy to [39]. In (16), $y = (y_1,\ldots,y_M)$ with $y_k \in [0,1]$ denotes the fraction of population being in infection state $k \in \{1,\ldots,M\}$, and the opinion state $x \in \mathbb{O}$ is "only" of dimension d. That is, equation (16) is a PDE for a density ρ that depends on $M+d$ dimensions. We have $\int \rho(x,y,t)dxdy = 1$ for appropriate initial conditions.

Remark 4.2. The co-evolution mean-field model (16) can easily be generalized to include further stochastic effects, e.g., by building it on the SPDE system introduced in Remark 3.3 instead of on the mean-field PDE in the opinion dimensions, or by integrating the chemical Fokker-Planck mean-field (9) model for the infection dimension.

4.4.1 Expectation values of observables

For later use it is important to note that the mean values of observables $\mathscr{A} = \mathscr{A}(x,y,t)$ of interest are given by

$$\langle \mathscr{A} \rangle(t) \;=\; \int \mathscr{A}(x,y,t)\rho(t,x,y)dxdy,$$

where $\rho = \rho(x,y,t)$ denotes the full solution of (16). If the required observable solely depends on one of the coordinates, say x, then the expression for the expectation values simplifies to

$$\langle \mathscr{A} \rangle(t) \;=\; \int \mathscr{A}(x,t)\hat{\rho}^y(x,t)dx,$$

where the marginal density is given by $\hat{\rho}^y(t,x) = \int \rho(t,x,y)dy$.

4.4.2 Fast opinion dynamics and slow infection dynamics

Before going into details of the dynamical properties of eq. (16), let us consider the limit of fast-scale opinion dynamics (timescale $\mathscr{O}(\epsilon)$) and slow infection dynamics (timescale $\mathscr{O}(1)$) in the sense that eq. (16) takes the form

$$\partial_t \rho^\epsilon = (\frac{1}{\epsilon} L_1 + L_2)\rho^\epsilon,$$

with $L_2\rho(x,y,t) = -\mathrm{div}_y(b(x,y)\cdot\rho(x,y,t))$. Following [24], Appendix A.5.3, on timescales $\mathscr{O}(1)$ the solution has the form

$$\rho^\epsilon(x,y,t) = \rho(y,t)\mu(x|y) + \mathscr{O}(\epsilon),$$

where $\mu(x|y)$ denotes the invariant density of L_1 that is reached from the initial conditions (exponentially quickly if the invariant density is unique and L_1 exhibits a spectral gap) on timescale $\mathscr{O}(\epsilon)$ via $\exp(-tL_1/\epsilon)$. For the simple case $W = 0$ there is a unique invariant density of L_1, such that

$$\mu(x|y) = \frac{1}{Z(y)} \exp\Big(-\frac{2}{\sigma^2}V(x,y)\Big),$$
$$Z(y) = \int \exp\Big(-\frac{2}{\sigma^2}V(x,y)\Big)dx.$$

In this case, we obtain

$$\partial_t \rho(y,t) = -\mathrm{div}_y(\bar{b}(y)\cdot\rho(y,t))$$

with

$$\bar{b}(y) = \int b(x,y)\mu(x|y)dx,$$

that is, transport along the trajectories of the averaged ODE system

$$\frac{dy}{dt}(t) = \bar{b}(y).$$

For $W = W(x,y) \neq 0$, the invariant density satisfies

$$\mu(x|y) = \frac{1}{\hat{Z}(y)} \exp\Big(-\frac{2}{\sigma^2}[V(x,y) + (W(\cdot,y) * \mu(\cdot|y))(x)]\Big),$$

where the convolution is only in x, and, in general, several solutions of the nonlinear equation exist.

Example 4.3 (SIR example continued)**.** For the SIR example with $y = (s,i,r)$, when we assume that $k_I = k_I(x)$ and k_R is an independent constant in x, the averaged form becomes:

$$\frac{ds}{dt}(t) = -\bar{k}_I(s,i,r)s(t)i(t)$$
$$\frac{di}{dt}(t) = \bar{k}_I(s,i,r)s(t)i(t) - k_R i(t)$$
$$\frac{dr}{dt}(t) = k_R i(t),$$

with $\bar{k}_I(s,i,r) = \int k_I(x,s,i,r)\mu(x|s,i,r)dx$, and initial values to be sampled from $\rho(s,i.r,t=0)$. Therefore, this is an ODE system with random initial conditions. For $W = 0$ we have

$$\bar{k}_I(s,i,r) = \frac{1}{Z(s,i,r)} \int k_I(x,s,i,r) \exp\Big(-\frac{2}{\sigma^2}V(x,s,i,r)\Big)\, dx.$$

5. Numerical methods

We first shortly review some appropriate methods for simulation of the co-evolution ABMs and then go into much more details concerning simulation of the co-evolution mean field equations, especially for continuous opinion space.

5.1. Co-evolution ABMs

The ABMs introduced in Sections 2.1, 3.1, and 4.1 all have the general form of Markov jump processes. Such processes can numerically be simulated by means of the well-known *Stochastic Simulation Algorithm* (SSA), as introduced by Gillespie [40].

The coupled ABM process of Sec. 4.3, however, is a hybrid model of a Markov jump process and a SDE. Numerical schemes have to couple a SDE-solver, e.g. an Euler–Maruyama discretization, with a SSA algorithm for the time-continuous infection part. The literature contains several scheme for this tasks: The first (and mostly simplest but numerically inefficient) option are *Synchronous time-discrete updating* schemes which are based on making a fixed (and very small) timestep Δt for each dimension independently and repeat after

coupling. This simulation scheme gives a good approximation of the true process only for Δt approaching zero [41]. Alternatives are given by so-called *Event-driven simulation* schemes [42] which first perform a SSA-step for the infection part with integrating hazard rates until a jump occurs, followed by steps of the SDE solver with updated and frozen infection states for the jump time. Or by the related *Temporal Gillespie* approach, see [43] for details.

5.2. Co-evolution mean field equations

The co-evolution mean-field PDE for discrete opinion space, equation (14), being a classical Liouville equation, can be solved by transporting the initial density along the trajectories of the associated ODE system. This is possible be means of applying a classical numerical solver for stiff ODEs for which many efficient implementations are available.

The co-evolution model with continuous opinion space, equation (16), is not discussed in comparable depth in the numerical literature, having the form of a non-linear PDE in $M + d$-dimensional continuous (x, y)-space. One option is to apply PDE solution techniques, e.g., Rothe-methods with a linear-implicit time stepping coupled to finite volume or finite element discretizations in (x, y)-space. This procedure, however, becomes very quickly inefficient or infeasible for $M + d > 3$. Next, we will present an alternative method based on particle schemes and operator splitting techniques.

5.2.1 Operator splitting

Given (16), a particularly easy way towards numerical solution leads to operator splitting techniques, e.g., Lie-Trotter splitting (first order convergent) [44] or Strang splitting (second order) [45]. For the sake of simplicity we will explain the procedure for Lie-Trotter splitting; generalizations, e.g., to Strang splitting should be obvious. To this end, we write equation (16) in the form $\partial_t \rho = (L_1 + L_2)\rho$ with

$$L_1 \rho(x, y, t) = -\mathrm{div}_y(b(x, y) \cdot \rho(x, y, t)),$$

choose a small timestep Δt and approximate the full solution $\rho(t + \Delta t)$ of $\partial_t \rho = (L_1 + L_2)\rho$ with initial condition $\rho(t) = \rho_*$ as initial density by

$$\rho(t+\Delta t) = \exp(\Delta t L_2)\ \exp(\Delta t L_1)\ \rho_* \ + \ \mathscr{O}(\Delta t^2). \tag{17}$$

In our case, this means that the first step can be computed by transport along the trajectories of the infection ODE (4) only in y,

$$\begin{aligned} \rho_{1/2}(x, y, t + \Delta t) &= \exp(\Delta t L_1)\ \rho_*(x, y) \\ &= \rho_*\Big(\Phi^{-\Delta t}(x, y)\Big) \\ &= \rho_*\Big(\Phi_x^{-\Delta t}(y)\Big), \end{aligned} \tag{18}$$

where the notation Φ_x^s indicates that the ODE system is solved solely in y but depends parametrically on x, that is, the coefficients of the ODE system for y depend on x. The second step,

$$\begin{aligned} &\rho(x, y, t + \Delta t) = \\ &\quad \exp(\Delta t L_2)\ \rho_{1/2}(x, y, t + \Delta t) \ + \ \mathscr{O}(\Delta t^2), \end{aligned} \tag{19}$$

means that the (numerical) solution of the mean-field PDE in x, that is, only in opinion space, has to be computed for the time interval $[0, \Delta t]$, starting with $\rho_{1/2}$, this time parametrized by y. In this step, the stepsize of the numerical solution of the PDE can be an integral fraction of Δt, i.e., it may involve several timesteps of the PDE solver.

5.2.2 Particle method based on Lie-Trotter splitting

The proposed scheme is a particle method for the co-evolution PDE (16). For the sake of simplicity we will present it for the case where the external potential V in the opinion dynamics part of the model is assumed to depend *only* on an expectation value $A_0(t) = \langle \mathscr{A}_0 \rangle(t)$ of an observable $\mathscr{A}_0 = \mathscr{A}_0(y)$, and *not* on y itself, that is,

$$V(x, y) = V(x, A_0).$$

Additionally, we assume that W does not depend on y at all.

Algorithm: First, we choose sampling points $(x^j(0), y^j(0))$, $j = 1, \ldots, n$ for the given initial

density $\rho(t = 0, x, y) = \rho_*(x, y)$ such that for "not too small" sets $A \subset \mathbb{O} \times \mathbb{Y}$

$$\int_A \rho_*(x, y) dy dy \approx \frac{1}{n} \sum_{j=1}^{n} \chi_A(x^j(0), y^j(0)),$$

where χ_A denotes the indicator function for the set A. Each "particle" centered at $(x^j(0), y^j(0))$ does not represent an individual agent but represents a sampling point of the overall mean-field density.

The proposed algorithm iteratively runs through the following sequence of steps, starting with $t = 0$ and temporal stepsize Δt:

1. We start at time t with n sampling points $(x_j(t), y^j(t)), j =, \ldots, n$.
2. Infection propagation step eq. (18): for each sampling point, new y-coordinates are computed:
$$y^j(t + \Delta t) = \Phi^{\Delta t}_{x_j(t)} y^j(t),$$
where we use the notation for the flow Φ^s_x of the associated ODE system and x-dependent $k_I(x)$, as introduced above.
3. Compute the update of the expectation value $A_0(t + \Delta t)$, that is, set
$$A_0(t + \Delta t) = \frac{1}{n} \sum_{j=0}^{n} \mathscr{A}_0(y^j(t + \Delta t)).$$
4. Opinion propagation step I eq. (19): compute the solution of the PDE
$$\partial_s \hat{\rho}_s = \operatorname{div}_x\Big(\hat{\rho}_s \cdot (\nabla_x V(\cdot, A_0(t + \Delta t)) + \nabla_x W * \hat{\rho}_s)\Big) + \frac{\sigma^2}{2} \Delta_x \hat{\rho}_s. \tag{20}$$
by means of a suitable discretization scheme with stepsize δt, an integer fraction of Δt, an appropriate uniform grid in the opinion space $\mathbb{O}$, and initial condition $\hat{\rho}_t = \hat{\rho}^y(t, \cdot)$. Then set $\hat{\rho}^y(t + \Delta t, \cdot) = \hat{\rho}_{t+\Delta t}$.
5. Opinion propagation step II: for each sampling point $(x^j(t), y^j(t + \Delta t))$ compute a new x-coordinate via the solution of the mean-field SDE:
$$dX_s = -\nabla V(X_s, A_0(t + \Delta t)) ds - (\nabla W * \hat{\rho}^y(t + \Delta t, \cdot))(X_s)\, dt + \sigma dB_s,$$
with initial condition $X(t) = x^j(t)$, and set $x^j(t + \Delta t) = X(t + \Delta t)$.
6. Compute all wanted approximations of mean values,
$$\langle \mathscr{A} \rangle(t + \Delta t) = \frac{1}{n} \sum_{j=0}^{n} \mathscr{A}(x^j(t + \Delta t), y^j(t + \Delta t), t + \Delta t).$$
7. Set $t \mapsto t + \Delta t$ and repeat.

Remark. If $W = 0$, the solution of the PDE in step 4 is not required and the mean-field SDE is a standard gradient-driven SDE. In this case, we get a pure particle method without PDE component. In principle, the whole scheme could in principle be repeated without the PDE computation (even if $W \neq 0$) by using enough particles so that the empirical density generated on x-dimension can be taken as a reliable sampling of the density. In this form, it can also be generalized to the general case of (16).

6. Numerical experiments

6.1. Co-evolution with discrete opinion space

We will consider several simple examples that illustrate the dynamical effect of opinion dynamics on infection dynamics and vice versa. To this end, we will utilize the simple 5-dimensional example introduced in Example 4.1.

Example 6.1 (Binary opinion space and SIR infection dynamics continued: ABM)**.** We continue with the ABM introduced in Example 4.1 with 8 transition rules and population space $\tilde{y}_1, \ldots, \tilde{y}_5$ denoting the numbers of agents of types S_0, I_0, S_1, I_1, R. We again use the notation $s_0 = \tilde{y}_1/N$, $i_0 = \tilde{y}_2/N$, $s_1 = \tilde{y}_3/N, i_1 = \tilde{y}_4/N$, and $r = \tilde{y}_5/N$, for the respective fractions. Then, the fraction of unconcerned and concerned in the population is

$$\rho_0 = s_0 + i_0 + r, \quad \rho_1 = s_1 + i_1,$$

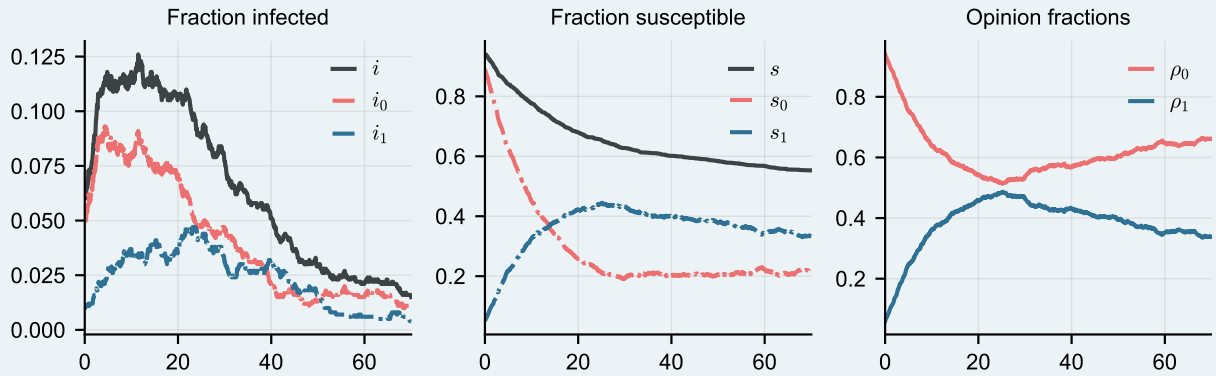

Figure 2: Realization of the co-evolution ABM ($N = 1000$) with binary opinion space and SIR dynamics outlined in Example 6.1. Left: Total fraction of infected for the pure SIR-ABM (without opinions) compared to i and its parts for the unconcerned and concerned, i_0 and i_1, in the co-evolution ABM. Middle: Total fraction of susceptibles s and its parts s_0 and s_1. Left: Fraction of concerned (ρ_0) and unconcerned (ρ_1).

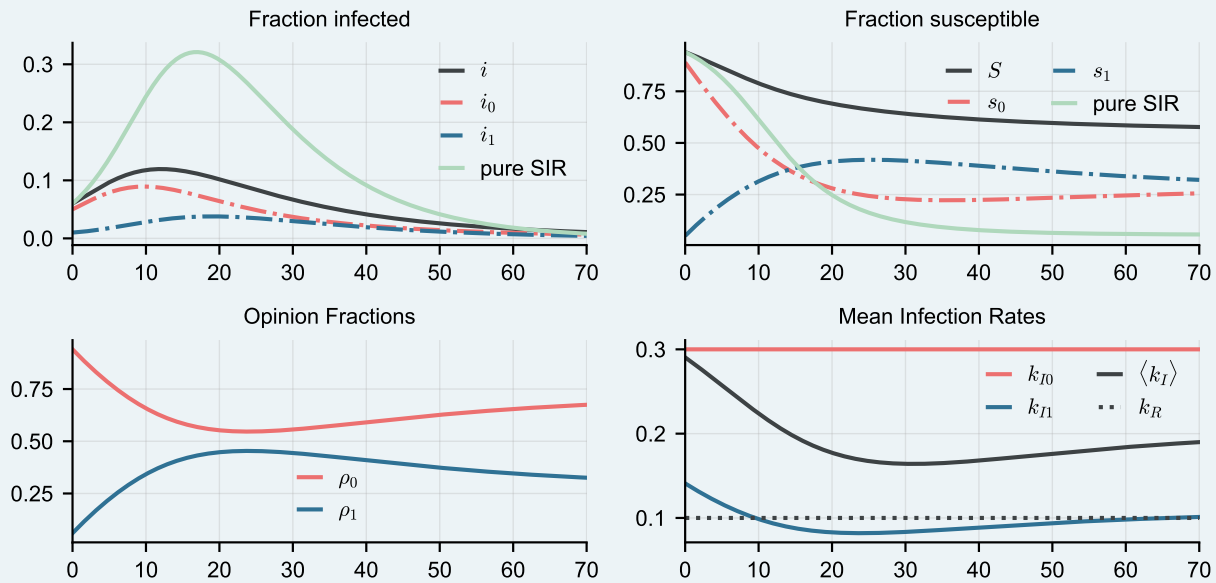

Figure 3: Coevolution with binary opinion space and SIR dynamics: 5-dimensional model with parameters outlined in Example 6.2. Top left: Fraction of infected for the pure SIR model (without opinions) compared to the total fraction of infected i and its parts for the unconcerned and concerned, i_0 and i_1, in the co-evolution model. Top right: Fraction of susceptible persons for the pure SIR model compared to the total fraction of susceptibles s and its parts s_0 and s_1 in the co-evolution model. Bottom left: Fraction of concerned (ρ_0) and unconcerned (ρ_1). Bottom right: Mean infection rate $\langle k_I \rangle$ and its parts k_{I0} and k_{I1} in comparison to the constant recovery rate k_R for the co-evolution model.

while the fraction of unconcerned that actively participate in infection spreading is given by $\hat{\rho}_0 = s_0 + i_0$, and the total fraction of infected–whether concerned or unconcerned—is given by $i = i_0 + i_1$. Moreover, we have to specify the kinetic rates k_{I0}, k_{I1}, k_R, a_0 and a_1. We set $k_{R1} = k_R = k_{R0}$, being constant at all times, and choose the basic SIR kinetic rates of the infection as introduced in Example 2.3 with $\tau = 10$ and $\mathcal{R}_0 = 3$:

$$k_R = 1/\tau_R = 0.1, \quad \text{and} \quad k_I = k_R \mathcal{R}_0 = 0.3.$$

Concerning the infection coefficients, we set $k_{I0} = k_I$, i.e., the basic SIR infection rate holds for the people be-

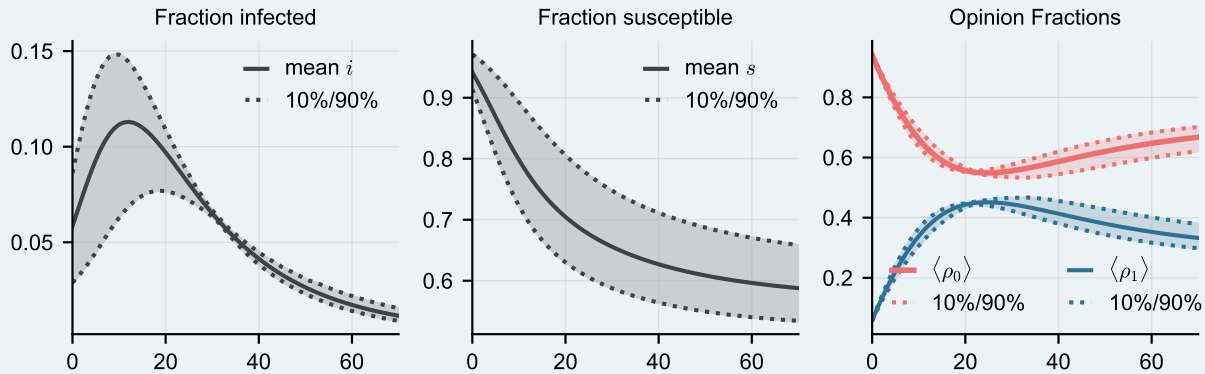

Figure 4: Running statistics of co-evolution density $\rho(x, y, t)$ for the binary opinion space and SIR dynamics: Left: Total fraction of infected (mean $\langle i(t)\rangle = \int (y_2(t) + y_4(t))\rho(x, y, t)dxdy$ and related quantiles for 10 and 90 percent). Middle: Total fraction of susceptible (mean $\langle s\rangle$ and quantiles). Right: Total fraction of unconcerned $\langle\rho_0\rangle$ and concerned $\langle\rho_1\rangle$ (mean and quantiles for each).

ing unconcerned and taking no special care for protection against being infected. Moreover,

$$k_{I1} = 0.5\, k_I\, (1 - \rho_1),$$

that is, the concerned, being careful and participating in social distancing, are subject to a reduced infection rate that additionally decreases as more agents get concerned. Thus, while the rate $k_{I0} = 0.3$ is constant in time, the rate k_{I1} changes with time and ρ_1. Additionally, we assume that the coefficients for opinion switching depend on the total fraction of infected i,

$$a_0 = \gamma_0(1 + 2i)^2, \quad a_1 = \gamma_1(1 - 2i)^2,$$

with $\gamma_0 = \gamma_1 = 1/30$. A realization of the associated co-evolution ABM, based on the SSA algorithm, is illustrated in Fig. 2.

Example 6.2 (Binary opinion space and SIR infection dynamics continued: Mean-field)**.** Next we consider the same setting as discussed in Example 6.1 for the associated mean-field model as of Example 4.1. Additionally, we introduce the *mean infection rate* in the population which is given by

$$\langle k_I\rangle = \frac{k_{I0}\hat{\rho}_0 + k_{I1}\rho_1}{\hat{\rho}_0 + \rho_1}.$$

The solution of the ODE system of the mean-field model (15) for initial states $i_0(0) = 0.05$, $i_1(0) = 0.01$, $s_1(0) = 0.05$, $s_0(0) = 1 - i_0(0) - i_1(0) - s_1(0)$, and $r(0) = 0$ is shown in Fig. 3. In the beginning, almost everyone is unconcerned until the infection spreads and people get concerned which dampens the mean infection rate (see bottom right panel). Through this process the infection peak is reached earlier and is three-fold lower than in the pure SIR model (without any opinion dynamics), see top left panel. The infection rate of the concerned, k_{I1}, for a period even drops below the recovery rate (see bottom right panel), slowing down infection spread under the concerned. Furthermore, by comparing the evolution of the co-evolution ABM for $N = 1000$, Fig. 2, with the associated mean-field model, Fig. 3, we observe good agreement of the typical dynamical behavior.

Additionally, Fig. 4 illustrates the solution $\rho(x, y, t)$ of the mean-field PDE system (14) for a normally distributed initial density $\rho(x, y, 0)$ with means given by the initial conditions used above and small variances $\sigma_{I0}^2 = 0.02$, and $\sigma_{I1}^2 = 0.002$,

$$\rho(x,y,0) \propto \exp\Big(-\frac{(i_0 - 0.05)^2}{2\sigma_{I0}^2}\Big)\exp\Big(-\frac{(i_1 - 0.01)^2}{2\sigma_{I1}^2}\Big) \times\delta(s_1 = 0.05)\delta(s_0 = 1 - i_0 - i_1 - s_1)\delta(r = 0).$$

Example 6.3 (Binary opinion space and double SIR infection dynamics)**.** We consider the case in which, in the setting of the previous example, a new virus variant enters the infection landscape at $t = 25$ with 3 infected persons. The new variant has an increased basic

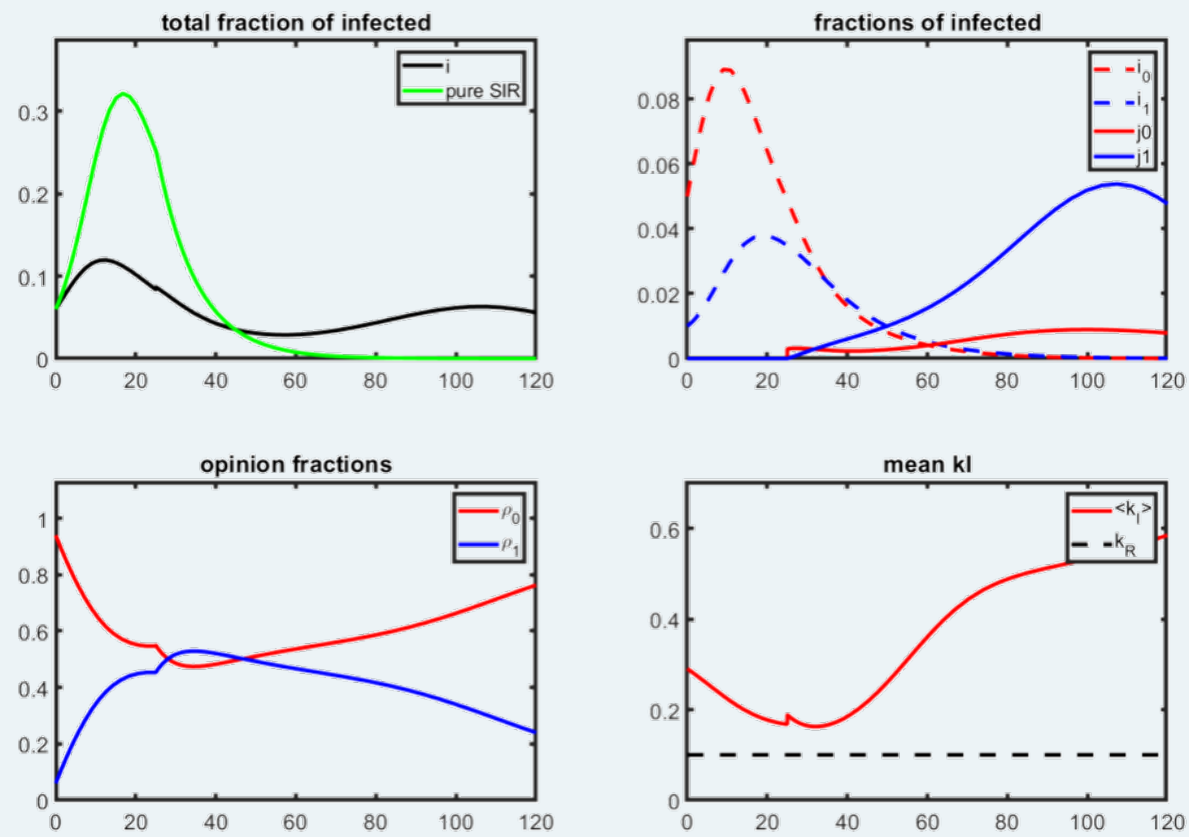

Figure 5: Coevolution with binary opinion space and double SIR dynamics: 7-dimensional model with parameters outlined in Example 6.3. Top left: Fraction of infected for the pure SIR model (without opinions) compared to the total fraction of infected i in the co-evolution model. Top right: Fractions of infected persons i_0, i_1, j_0 and j_1 in the co-evolution model. Bottom left: Fraction of concerned (ρ_0) and unconcerned (ρ_1). Bottom right: Mean infection rate $\langle k_I \rangle$ for the co-evolution model.

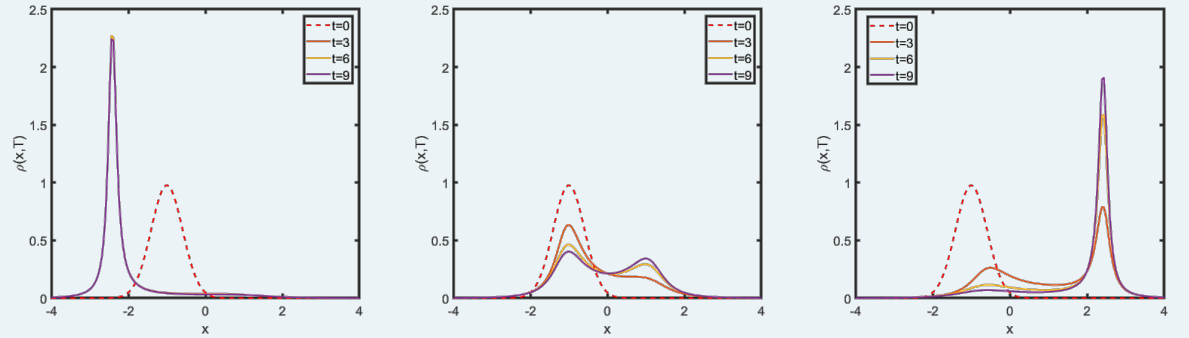

Figure 6: Evolution of the opinion density $\rho(x,t)$ in the mean-field opinion model with external double well potential $V(x,I) = (x^2 - 1)^2 - 8/3\phi(I)x^3$ for $\phi(I) = -\frac{8}{3}I^2 + \frac{14}{3}I - 1$, $\sigma = 1$, $W = 0$, and $t = 0, 3, 6, 9$. Left: $I = 0$. Middle: $I = 1/4$. Right: $I = 1$.

infection rate of $k_J = 4k_I$ but the same recovery rate, and people having recovered from the first variant are immune against the second. The fraction of persons being infected with the second variant are denoted j_0 and j_1, again for unconcerned and concerned. Hence, the evolution of the associated mean-field PDE (14) occurs by transport along the trajectories of a 7-dimensional ODE system, with states $s_0, s_1, i_0, i_1, j_0, j_1, r$. The ODE reads

$$
\begin{aligned}
\frac{ds_0}{dt}(t) &= -k_{I0}(x)s_0(t)i_0(t) - k_{J0}(x)s_0(t)j_0(t) - a_0(y)s_0 + a_1(y)s_1 \\
\frac{di_0}{dt}(t) &= k_{I0}(x)s_0(t)i_0(t) - k_{R0}i_0(t) - a_0(y)i_0 + a_1(y)i_1 \\
\frac{dj_0}{dt}(t) &= k_{J0}(x)s_0(t)j_0(t) - k_R j_0(t) - a_0(y)j_0 + a_1(y)j_1 \\
\frac{ds_1}{dt}(t) &= -k_{I1}(x)s_1(t)i_1(t) - k_{J1}(x)s_1(t)j_1(t) + a_0(y)s_0 - a_1(y)s_1 \\
\frac{dj_1}{dt}(t) &= {}_{J1}\ (x)s_1(t)j_1(t) - k_R j_1(t) + a_0(y)j_0 - a_1(y)j_1 \\
\frac{di_1}{dt}(t) &= k_{I1}(x)s_1(t)i_1(t) - k_{R1}i_1(t) + a_0(y)i_0 - a_1(y)i_1 \\
\frac{dr}{dt}(t) &= k_R(i_0(t) + i_1(t) + j_0(t) + j_1(t),
\end{aligned}
$$

where in addition to the rate parameters of the previous example, we set $k_{J0} = k_J$, and $k_{J1} = 0.5\, k_J\, (1 - \rho_1)$.

We also assume that for $t \geq 25$ the rate of switching opinions to be concerned is 3-fold as for $t < 25$, in order to take the real-world observation into account that people get concerned faster after a first infection wave. In the new scenario, the opinion fractions are $\rho_0 = s_0 + i_0 + j_0 + r$ and $\rho_1 = s_1 + i_1 + j_1$. while the total fraction of infected is given by $i = i_0 + i_1 + j_0 + j_1$.

Fig. 5 illustrates the evolution of this double infection scenario, comparing the pure (4-dim) SIR model and the (7-dim) co-evolution model. We observe that the co-evolution model results in a prolongation of the infection period with a second wave of infections after the appearance of the second variant and significantly reduced maximal infection spread, while the pure SIR model shows a single, more intense initial infection wave where a large part of the population gets infected and is no longer subject to the new form of the infection. We also observe that the upcoming new variant increases the fraction of concerned, but only shortly since the limited total infection rates stay relatively low.

6.2. Co-evolution with continuous opinion space

In order to preform numerical experiments for the co-evolution model (16), we apply the particle method introduced in Sec. 5.2.2. To this end, we assume that the infection part of the model is given by SIR-model with variables $y = (y_1, y_2, y_3) = (s, i, r)$ and the opinion part is modeled with 1-dimensional continuous opinion space $\mathbb{O} = [-3, 3]$ that ranges from being unconcerned for $x < 0$ to concerned for $x > 0$.

6.2.1 Setting

For the SIR part, we consider the scaled propensities $\tilde{\alpha}_1(x, y) = k_I(x)y_1y_2$ and $\tilde{\alpha}_2(x, y) = k_R y_2$, where k_R is independent of the opinion component x. Regarding the calibration of k_I on x, we assume $\tau_R = 10$ days, i.e., $k_R = 0.1$ per day. Furthermore, for the totally unconcerned around $x = -2$, we set $\mathcal{R}_0 \approx 10k_{I,max}$, and for the concerned around $x = 2$ we set $\mathcal{R}_0 < 1$ so that the infection is no longer spreading but decreasing if everyone would be concerned. That is, for the unconcerned we have

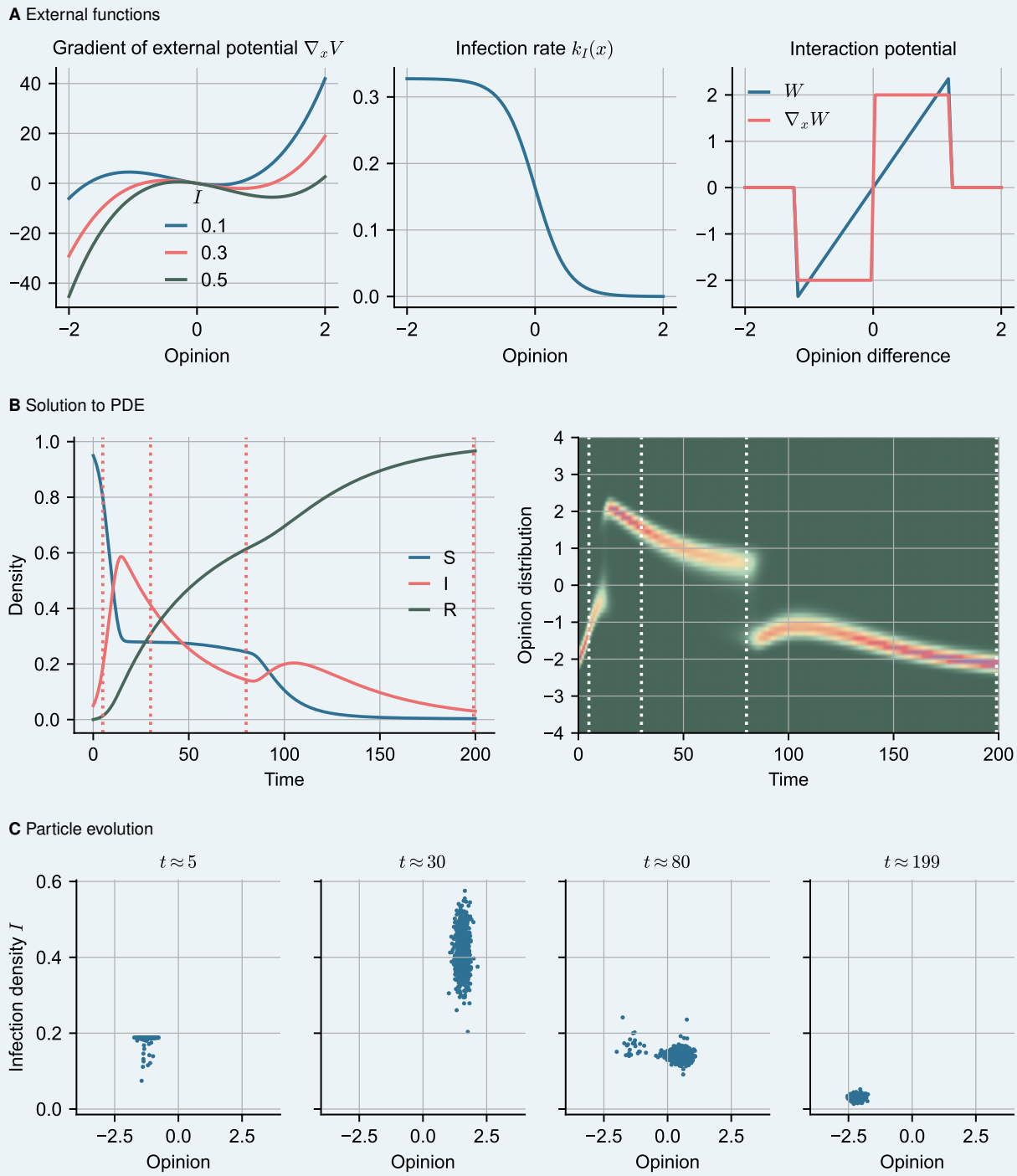

Figure 7: Modelling coevolving infections and continuous opinions. **A** The functions driving the dynamics: external potential $\nabla_x V$ with $I_{crit} = 0.25$, infection rate k_I with $k_{min} = 0.0225$ and $k_{max} = 0.35$; and the bounded confidence interaction potential with $R = 1.5$ and $\kappa = 5$. **B** The solution to the coupled PDE eq. (16) using the particle method. Also indicated are four time values at which we evaluate the particle distribution in infection-opinion space (**C**).

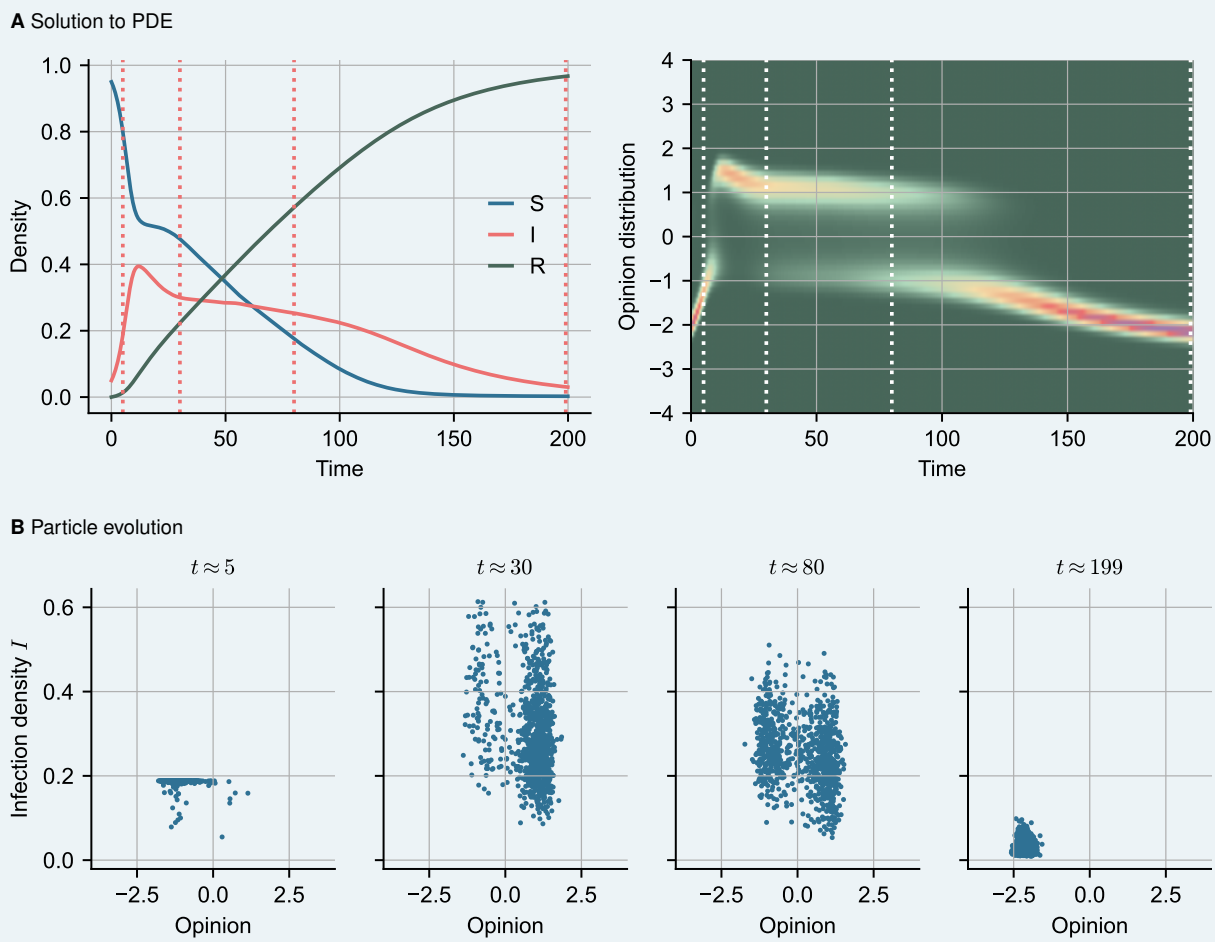

Figure 8: The same figures as in Fig. 7 (B) and (C) but with no opinion interaction, i.e. $\kappa = 0$.

$k_I(x = -2) \approx \mathcal{R}_0 \cdot 0.1 = k_{I,max}$ per day and for the concerned $k_I(x = 2) \approx 0.9 \cdot 0.1 = 0.09 = k_{I,min}$, for example. For the function $k_I = k_I(x)$ we propose

$$k_I(x) = -\frac{1}{2}(k_{I,max} - k_{I,min}) \cdot \tanh(2x) + \frac{1}{2}(k_{I,max} + k_{I,min}),$$

such that approximately $k_I(x = -2) \approx k_{I,max}$,$k_I(2) \approx k_{I,min}$ and $k_I(0) = (k_{I,max} + k_{I,min})/2$.

Regarding the coefficients of the opinion part, we propose that the assumptions made in Sec. 5.2.2 are matched: The external potential V in the opinion dynamics part of the model is assumed to depend *only* on the expectation value wrt ρ of the infection observable $y_2 = i$, i.e., on the total fraction of infected,

$$I(t) = \int y_2 \rho(x, y, t)\, dy\, dx,$$

and not on y in detail, $V(x, y) = V(x, I)$. We assume that the external potential V has the form of a double well potential given by

$$V(x, I) = (x^2 - 1)^2 - \frac{8}{3} a\, \phi(I)\, x^3 + \text{drift} \cdot x, \quad (21)$$

where $a > 0$ and

$$\phi(I) = \alpha I^2 + (2 - \alpha) I - 1, \quad \alpha = \frac{1 - 2I_{crit}}{I_{crit}^2 - I_{crit}}$$

gives a dependence on a given value $I \in [0, 1]$, and predefined value $I_{crit} \in (0, 1)$. In the following, we set

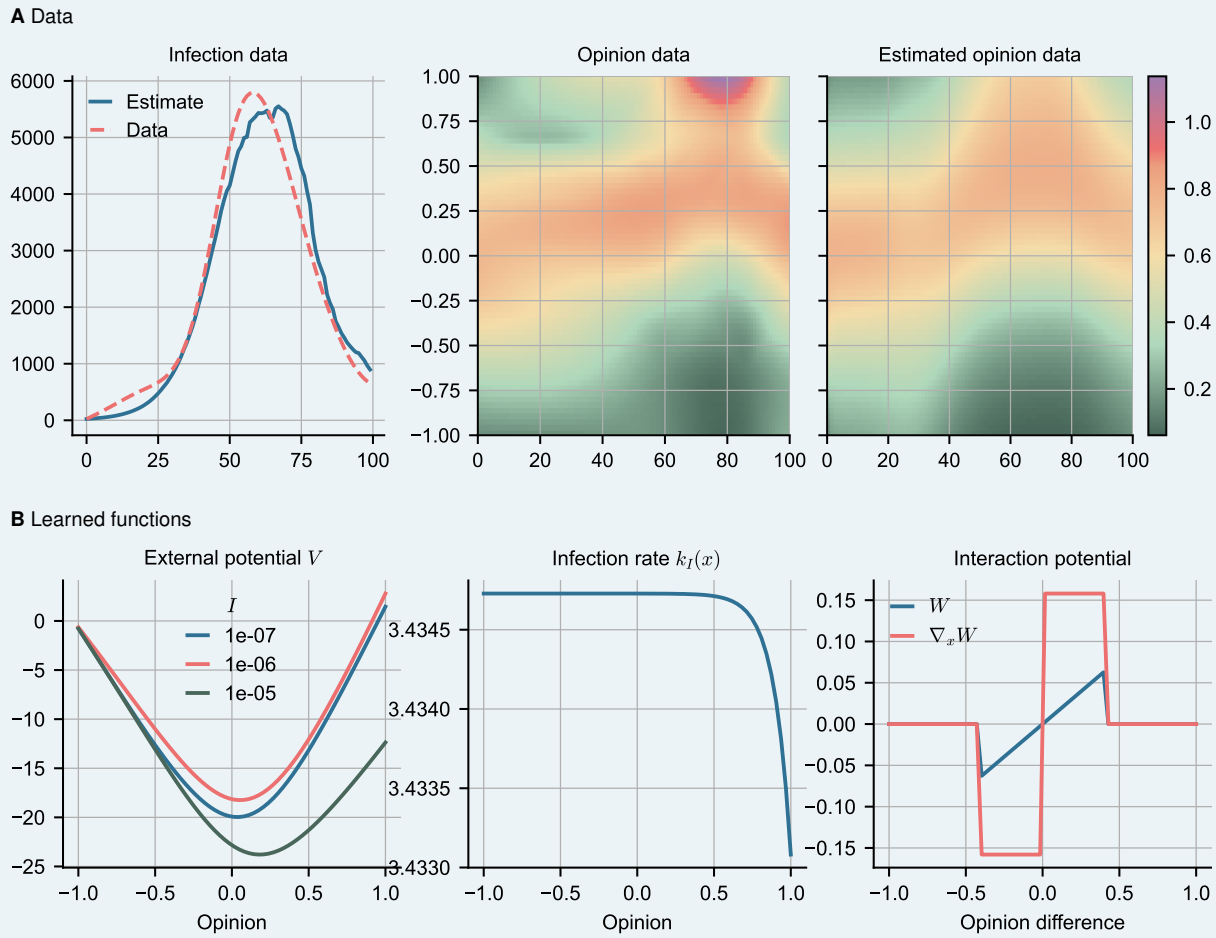

Figure 9: Calibrating the model eq. (16) to infection-opinion data from Japan for early 2020. All functions and parameters are as in fig. 7 with the exception of the external potential $V(x, I)$, which is modelled as a neural network. **A** Left: the infection data (red) and the calibrated model estimates (blue); center: the smoothed anxiety levels taken from the survey data [46]; right: the PDE estimate. **B** Left: the learned interaction potential V, parametrised as a deep neural network. Centre: the learned infection rate $k_I(x)$; right: the bounded confidence interaction potential W.

$a = 1$, $I_{crit} = 1/4$, and drift$= 0$; changing the value of drift gives more options for asymmetry between the two wells of the potential.

Assuming that 1 unit in time is 1 day like in the SIR part, we want the equilibration timescale for the double well potential for the balance situation ($I = I_{crit}$) to be about $e^2 \approx 7$ days. In this situation the potential barrier is $\Delta V = 1$ with equilibration time roughly $\exp(\beta \Delta V)$ and $\beta = 2/\sigma^2$, where σ is the noise intensity. Thus, we have to choose approximately $\sigma = 1$. In order to understand the typical dynamics originating from the external potential, we first ignore the infection part and study the solution of the opinion dynamics mean-field PDE (20) for this setting for $W = 0$, see Fig. 6.

As interaction potential W, we choose as in a standard bounded confidence interaction such that

$$(\nabla_x W * \rho_t)(x) = \kappa \int_{\mathbb{R}} \mathbb{1}_{|x'-x| \leq R} \cdot \rho_t(x') dx', \quad (22)$$

where $\kappa > 0$ is the interaction strength, $R > 0$ the interaction radius and $\mathbb{1}$ denotes the indicator function. That is, W is assumed to be independent of the infection component. This corresponds to a commonly used interaction potential between agents which interact with each other if their opinions are similar enough, see e.g. the Hegselmann-Krause model [29].

6.2.2 Numerical experiments

As initial condition for the co-evolution model eq. (16), we use

$$\rho(t=0,x,y) = \frac{1}{Z}\exp(-3(x+1)^2)\times \\ \delta(y_1 = 0.95)\delta(y_2 = 0.05)\delta(y_3 = 0),$$

that is, the initial fraction of infected is 0.05 and almost the entire population is unconcerned in the beginning.

Fig. 7 illustrates the external and interaction potential and the evolution of the coupled system for interaction strength $\kappa = 5$. Fig. 8 illustrates the same scenario for interaction strength $\kappa = 0$, i.e., without inter-agent opinion interaction. For $\kappa = 5$ we initially observe a stark increase in the fraction of concerned with increasing infection numbers. Then, with decreasing infection numbers, the level of concern drops while around $t = 80$ a sudden switch from being mildly concerned to being mildly unconcerned happens. This jump leads to a distinct second infection wave. The bottom part of Fig. 7 shows the distribution of particles in the (x, i)-projection of the configuration space as it results from the numerical scheme. For $\kappa = 0$, we neither observe a sudden jump from concerned back to unconcerned levels nor the second infection wave. The evolution is more smoothly going through a longer transient period between $t = 30$ and $t = 100$, where concerned and unconcerned opinions coincide.

7. Discussion

This work presents a coherent modeling framework for the coevolution of infection and opinion dynamics, bridging individual-based agent models with their continuum large-population limits. By coupling well-established models for epidemic spreading with discrete and continuous opinion dynamics, we demonstrate how social behavior and health status can interact and jointly shape the macroscopic evolution of a population. The modeling approach is intentionally modular and not tailored to any specific infection or opinion mechanism. Rather, it is designed as a systematic pathway for integrating and analyzing coevolution models based on arbitrary building blocks from epidemiology and social dynamics.

In addition to laying the theoretical foundation, we provide a discussion of appropriate numerical strategies for simulating these complex systems efficiently and reliably. Our numerical experiments illustrate how the interaction between social influence and infection dynamics can lead to a rich set of phenomena, including second infection waves, polarization-induced phenomena, and spatially localized behavioral effects. Such mechanisms underscore the importance of considering behavioral dynamics not just as an external input to epidemiological models but as a co-evolving system component.

The framework we propose supports a wide range of future developments. These include the integration of data-informed parameter estimation, sensitivity analysis under model uncertainty, and the simulation of intervention strategies such as targeted vaccination campaigns, public information policies, or mobility restrictions. Our approach provides the methodological flexibility to incorporate more realistic mobility, demography, and social structure, making it applicable to both stylized theoretical studies and real-world decision support.

In sum, this work contributes to a deeper understanding of the interplay between behavior and disease by offering a tractable and extensible modeling framework. It enables a new generation of co-evolution models that are capable of reflecting the feedback loops between individual decisions and epidemic evolution—an essential step toward designing robust, adaptive responses to public health challenges.

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Data availability All code and data is available at [LINK].

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Supporting Information

Supporting information goes here.